



# Yueju-Ganmaidazao Decoction confers rapid antidepressant-like effects and the involvement of suppression of NMDA/NO/cGMP signaling

Hailou Zhang<sup>a,c,1</sup>, Yan Sun<sup>a,1</sup>, Shiyu Qian<sup>a,1</sup>, Rui Ge<sup>a</sup>, Xiaoyan Guo<sup>a</sup>, Qinqin Shen<sup>a</sup>, Li Sheng<sup>a</sup>, Chunying Nie<sup>a</sup>, Yi Zhang<sup>a</sup>, Yao Yao<sup>a</sup>, Tong Zhou<sup>a</sup>, Wei Wang<sup>a</sup>, Wenda Xue<sup>a,c,\*\*</sup>, Gang Chen<sup>b,c,\*</sup>

<sup>a</sup> Key Laboratory of Integrative Biomedicine for Brain Diseases, Nanjing University of Chinese Medicine, Nanjing, 210023, China

<sup>b</sup> Interdisciplinary Institute for Personalized Medicine in Brain Disorders and Research Center for TCM Fang-Zheng, Jinan University, Guangzhou, 510632, China

<sup>c</sup> Co-innovation Center of Neuroregeneration, Nantong University, Nantong, Jiangsu, 226001, China

## ARTICLE INFO

### Keywords:

YG  
Antidepressant-like  
NMDA  
NO  
cGMP  
NR1

## ABSTRACT

**Ethnopharmacological relevance:** Yueju-Ganmaidazao Decoction (YG) is a multitherbal medicine prescribed for treatment of mood disorder, consisting of two classical traditional Chinese herbal medicine Yueju and Ganmaidazao. Yueju and Ganmaidazao both are used for depression treatment. The combined decoction of Yueju and Ganmaidazao is prescribed to achieve optimal clinical outcomes by dealing with different symptoms of depression. Recent studies indicated ethanol extract of Yueju was capable to confer rapid antidepressant-like response. The antidepressant activity of YG decoction with fast-onset feature remains to be investigated.

**Aim of the study:** Rapid and safe antidepressant treatment is urgently needed. This study aimed to assess the rapid antidepressant-like activity of YG and the underlying mechanism, focusing on NMDA/NO/cGMP signaling. **Materials and methods:** The optimal doses for immediate and persistent antidepressant-like response were first screened using tail suspension test (TST) and forced swimming test (FST) post a single administration of YG. The rapid action was further confirmed by using the chronic mild stress (CMS) and learned helplessness (LH) paradigms. The expressions of NMDA receptor subunits were evaluated post stress and YG. The contributions of NMDA, NO, and cGMP signaling to the antidepressant effect of YG were investigated systematically using pharmacological interventions.

**Results:** The optimal dose for immediate and persistent antidepressant potential, evidenced with reduced immobility times in TST or FST from 30 min to 7 days, was determined. The rapid antidepressant-like effect was confirmed in CMS and LH paradigms, including instant normalization of sucrose preference behavior. The expression of NMDA subunit NR1 in the hippocampus was reduced from 30 min to 5 days post YG. In animals subjected to CMS and LH, hippocampal NR1 expression increased, reversed by YG. YG's antidepressant-like effect was blunted by pretreatment with the agonists along the signalings including NMDA (75 mg/kg), L-arginine (750 mg/kg) and sildenafil (5 mg/kg) in TST or FST. Conversely, administration of subeffective dose of individual antagonists, including MK-801 (0.05 mg/kg), 7-nitroindazole (30 mg/kg), methylene blue (10 mg/kg), in combination with a subeffective dose of YG, elicited antidepressant effects.

**Conclusion:** YG conferred rapid antidepressant-like effects, and the antidepressant response was essentially dependent on suppression of NMDA/NO/cGMP signaling.

**Abbreviations:** YG, Yueju-Ganmaidazao Decoction; TST, Tail Suspension Test; FST, Forced Swimming Test; CMS, Chronic Mild Stress paradigm; LH, Learned Helplessness paradigm; SPT, Sucrose preference test; OFT, Open field test; NSF, Novelty-suppressed feeding test; NMDA, N-methyl-D-aspartic acid; NO, nitric oxide; cGMP, cyclic guanosine 3,5-monophosphate

\* Corresponding author. Interdisciplinary Institute for Personalized Medicine in Brain Disorders, and Formula-Syndromes Research Center, Jinan University, Guangzhou, 510632, China.

\*\* Corresponding author. Key Laboratory of Integrative Biomedicine for Brain Diseases, Nanjing University of Chinese Medicine, Nanjing, 210023, China.

E-mail addresses: [xuewenda@njucm.edu.cn](mailto:xuewenda@njucm.edu.cn) (W. Xue), [hdn\\_2001@yahoo.com](mailto:hdn_2001@yahoo.com), [chengang@jnu.edu.cn](mailto:chengang@jnu.edu.cn) (G. Chen).

<sup>1</sup> Contributed equally to the work.

<https://doi.org/10.1016/j.jep.2019.112380>

Received 27 April 2019; Received in revised form 24 October 2019; Accepted 6 November 2019

Available online 07 November 2019

0378-8741/ © 2019 Elsevier B.V. All rights reserved.

## 1. Introduction

Major depressive disorder (MDD), or depression, is one of most prevalent lifetime psychiatric disorders (Zomkowski et al., 2012). According to the World Health Organization, depression will become the most leading cause of diseases worldwide (Coumans et al., 2016). It takes several weeks for the conventional antidepressants, such as selective serotonin reuptake inhibitors (SSRIs), to reach efficacy, which can be very dangerous for patients at high risk of suicide (Li et al., 2010). Recently, ketamine, a selective N-methyl-D-aspartic acid (NMDA) receptor antagonist, has been shown to elicit rapid antidepressant effects both in patients and animal models of depression (Duman, 2018; Duman et al., 2012). The antidepressant effects start within 2 h and persist for several days after a single dose administration (Li et al., 2010). Lately, ketamine's enantiomer esketamine has been approved by the FDA of USA for prevention of suicide. However, the concerns for the potential adverse effects of ketamine such as addiction and neurotoxicity may restrict its clinical use (Bonnet, 2015). It is still urgent and necessary to develop fast-onset and safe antidepressants.

One strategy to discover the potential rapid and safe antidepressants is to explore alternative medicine such as Chinese herbal medicine, which has been widely used to alleviate the symptoms of emotional disorders including depression (Chen and Guo, 2017; Fan et al., 2018). Yueju pill and Ganmaidazao decoction are two of the mostly used and long-history formula to treat depression (Chen and Guo, 2017). Yueju consists of 5 indigenous medicine: the rhizoma of *Cyperus rotundus* L, the rhizoma of *Ligusticum chuanxiong* Hort, the fruit of *Gardenia jasminoides* Ellis, the rhizoma of *Atractylodes lancea* (Thunb) DC and *Massa fermentata*, which is a mixture of six kinds of substances produced by fermentation including *Prunus armeniaca* L. (Rosaceae), *Phaseolus angularis* Wight. (Leguminosae), *Artemisia annua* L. (Compositae), *Xanthium sibiricum* Patr. (Compositae) and *Polygonum hydropiper* Linn. (Polygonaceae). All of components except *Massa fermentata* are capable to elicit antidepressant activity (Wei et al., 2008). Recently, the ethanol extract of Yueju pill has been demonstrated to confer ketamine-like rapid antidepressant effects in various models including chronic mild stress (CMS) and learned helplessness (LH) (Tang et al., 2015; Xue et al., 2013, 2016), and facilitate the fast-onset antidepressant efficacy in a pilot clinical trial (Wu et al., 2015). Ganmaidazao decoction consists of *Curabitar triticum*, *Licorice root* and *Fructus Ziziphi Jujubae*, in which *Licorice* has multiple bioactive compounds with antidepressant potential (Zhao et al., 2008). A meta-analysis indicated Ganmaidazao decoction has few side effects and the potential as an antidepressant (Yeung et al., 2014). Yueju and Ganmaidazao formula were frequently prescribed jointly for better treatment outcomes, by simultaneously dealing with the different symptoms of depression including sleep-related disorder, anxiety, and depressed mood (Feng et al., 2016). The representative constituents of Yueju-Ganmaidazao includes Iridoids, Crocins, Quercetin, Ferulic acid,  $\alpha$ -cyperone,  $\beta$ -cyperone, ligustilide,  $\alpha$ -Aromatic ketone, *Atractylodesin*, *Glycyrrhizin*; *Glycyrrhizinate*, *atractylin* and *Aflatoxin* et al. (Supplementary Table 1). Although the antidepressant effects and mechanisms for each one have been reported in a few number of studies, those for YG remain to be assessed (Lou et al., 2010; Tang et al., 2015). Especially, it is intriguing whether the combination formula is still capable to elicit rapid antidepressant-like effects as Yueju does.

As indicated by ketamine, increasing number of studies support the important role of glutamate transmission in mood disorders and antidepressant action (Du et al., 2006; Tanti and Belzung, 2010). Some other NMDA receptor antagonists also show antidepressant-like effects (Amiri et al., 2016; Donello et al., 2018). Additionally, compounds that impede NMDA receptor delivery exhibit antidepressant-like effects (Amiri et al., 2016; Donello et al., 2018). Conversely, pretreatment of NMDA agonist abolishes the effect of ketamine (Burgdorf et al., 2013; Hillhouse and Porter, 2014). Yueju and ketamine both immediately and lastingly inhibit NMDA subunit NR1 expression, associated with their

immediate and persistent antidepressant activity (Tang et al., 2015). NMDA signaling regulates depression via several different pathways, including the NO-cGMP signaling. NMDA receptor activation stimulates generation of nitric oxide (NO), which increases the level of cyclic guanosine 3,5-monophosphate (cGMP) (Araujo et al., 2018; Neitz et al., 2014; Vuckovic et al., 2015). Reduction of the NO level induces antidepressant-like effect (Haj-Mirzaian et al., 2016a,b) and blockage of NO activity enhances the antidepressant activity of SSRIs and other agents (Hillhouse and Porter, 2014; Neitz et al., 2014; Vuckovic et al., 2015). Additionally, the level of cGMP has been shown to influence the antidepressant effect of some SSRIs (Hartell, 1996; Read et al., 2001).

The present study aimed to test the rapid antidepressant-like activity of YG and the underlying mechanisms, focusing on NMDA/NO/cGMP signaling. The doses to potentially induce rapid antidepressant-like action of YG were first to be screened. The rapid antidepressant-like action was further validated using two animal models of depression, learned helplessness and chronic mild stress. We investigated the impact of YG on the expression of NMDA receptor subunits in normal or stressed condition. Particularly, the role of NMDA/NO/cGMP signaling in the antidepressant effects of YG were systematically examined, using various antagonists/inhibitors or agonists.

## 2. Materials and methods

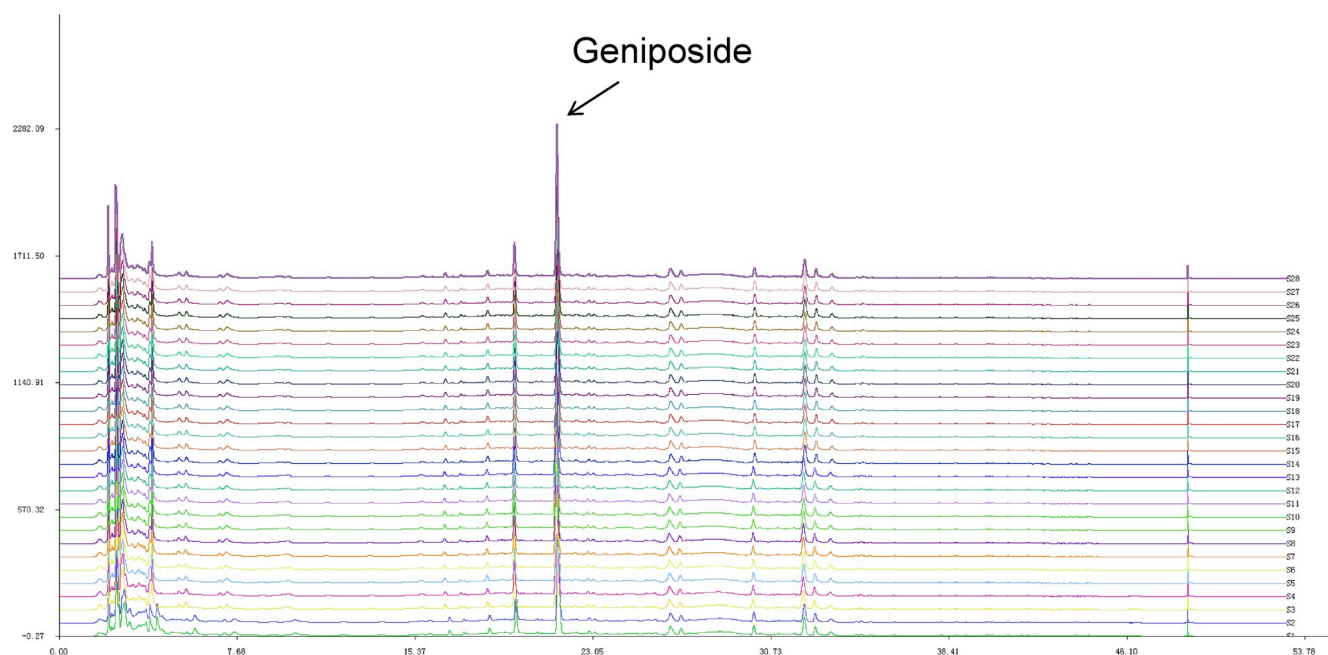
### 2.1. Animals

Kunming mice (23–25 g) were purchased from Shanghai Sippr-BK Laboratory Animal Co., Ltd. Mice aged 6 to 8 weeks were used to animal facilities for one week before the experiment. Animals were maintained in standard laboratory conditions (temperature:  $22 \pm 2$  °C, indoor humidity:  $50 \pm 10\%$ ; 3–5 animals per cage (290\*178\*160 mm)). The light/dark period was 12:12 h. These animals are fed standard food and filtered water. The Regulations for Animal Experiment conform to the Guidelines for the Nursing and Use of Laboratory Animals and are approved by the Animal Nursing and Use Committee of Nanjing University of Chinese Medicine.

### 2.2. Drugs

Experimental Chinese Medicinal herbs purchased from Nanjing University of Chinese Medicine Chinese Pharmacopoeia, the outpatient department of the School of Medicine, has been identified by experts from the College of Pharmacy of our University. Yueju-Ganmaidazao Decoction consists of *Cyperus rotundus* L (CR) 100 g, *Ligusticum chuanxiong* Hort(LC) 100 g, *Gardenia jasminoides* Ellis (GJ) 100 g, *Atractylodes lancea* (Thunb)DC (AL) 100 g, *Massa fermentata* (MF) 100 g, *Curabitar triticum* (CB)125 g, *Licoriceand* (LR)250 g, *Fructus Ziziphi Jujubae* (FZJ) 375 g. The above materials were soaked with water for 30 min in a 1:8 ratio and then heated for 1 h. Subsequently, the solution was filtered and collected. This procedure was repeated twice. All collected solution was placed in a water bath at 60 °C, and evaporated to a designated concentration. The yield of the extraction was 20%. YG was administrated via intragastric gavage (i.g.). The fingerprints of YG for quality control is shown in Fig. 1. The species and the main ingredients of each herb were shown in Supplemental Fig. 1. The dosage of the each drug followed Chinese Pharmacopoeia (version 2010).

L-NAME (10 mg/kg), a non-specific NOS inhibitor; 7-nitroindazole (30 mg/kg.), an nNOS specific inhibitor; L-Arginine (750 mg/kg), NO precursor; Methylene blue (10 mg/kg), an inhibitor of both NOS and sGC; NMDA (75 mg/kg), a NMDA receptor agonist; MK-801 (0.01 mg/kg), a NMDA receptor antagonist; Sildenafil (5 mg/kg), a phosphodiesterase 5 inhibitor, were purchased from Sigma, St. Louis, MO, USA. All drugs solution were made in saline (0.9%) except 7-nitroindazole that was dissolved in 5% Tween 80. All drugs were administered intraperitoneally (i.p.) with a volume of 10 ml/kg body weight. The



**Fig. 1.** The fingerprints of YG for quality control. Ten samples of YG were used to develop the standard fingerprints. Average chromatographic fingerprint obtained from the software “Similarity Evaluation System for Chromatographic Fingerprints”.

pretreatment time of agonist/antagonist or inhibitor before single dose of YG is slightly different (Supplemental Fig. 1).

HPLC finger was reported previously (Xue et al., 2013; Zhang et al., 2015). Briefly, ten batches of YG were analyzed by high performance liquid chromatography fingerprint analysis system. High performance liquid chromatography (HPLC) analysis was performed using a Waters 2695 Alliance HPLC system (Waters Corp., Milford, MA, USA) equipped with a quaternary ammonium salt pump solvent management system, an on-line degasser, and an autosampler. Raw data was detected using the 2998 DAD and processed using Empower software. An Apollo C18 column (250 mm × 4.6 mm, 5 μm) before the water symmetric shield RP was analyzed using a C18 guard queue (20 mm × 3.9 mm, 5 μm). The injection volume was 10 μl and the column temperature was maintained at 30 °C. The DAD detector was set at 265 nm to obtain a chromatogram. The mobile phase consisted of A (methanol) and B (0.1% aqueous acetic acid, v/v), gradient elution: 0–15 min, 10–20%A, 15–25 min, 20–55%A, 25–40 min, 55–90% A, 40–45 min, 90–100%A. The mobile phase flow rate was 1.0 ml/min. The HPLC-DAD chromatographic data of 10 test samples were analyzed using the professional software “Chinese Medicine Chromatographic Fingerprint Similarity Evaluation System” (2004 A version) to extract the average chromatogram and similarity.

## 2.3. Behavioral tests and animal models

### 2.3.1. Open field test (OFT)

The OFT assesses locomotor activity and anxiety-like behavior in bright open areas. The test was carried out in a well-lit (~300 lux) clear acrylic cage (40 × 40 × 15 cm). The mice were gently placed in the center and explored on the left for 5 min. A digital image of the path taken by each mouse was tracked with a camera, and the total running distance (locomotor activity) and time spent at the center were analyzed using the ANY-maze software. The test instrument was thoroughly cleaned with 70% ethanol and then dried between each animal.

### 2.3.2. Tail suspension test (TST)

In a soundproofed and visually isolated room, tape a single mouse about 1 cm from the tip of the tail to make it 50 cm above the ground. Animal activities were recorded. ANY-maze software (Stoelting Co.,

Ltd., USA) was used to calculate the total time of rest during the last 4 min of the 6 min test period (Thierry et al., 1984).

### 2.3.3. Forced swimming test (FST)

FST is also a widely used paradigm for behavior despair and antidepressant response in rodents (Thierry et al., 1984). The mice were removed from the home cage and placed in a clear glass jar (40 cm high, 20 cm diameter) filled with 30 cm of water (22–23 °C) and allowed to swim for 6 min. At the end of the test, the animals were removed from the water, wiped dry with a paper towel, and returned to the home cage. When mice floated in the water without struggling, they are considered to be immobile, and only those actions necessary to keep their heads above the water. The total immobility time was analyzed by ANY-maze software during the last 4 min of the 6 min test.

### 2.3.4. Learned helplessness (LH) model

Procedure for LH was followed previously (Maeng et al., 2008; Xue et al., 2013; Zhang et al., 2015). Briefly, mice were put in soundproofed two-way shuttle boxes (40 × 10 × 13 cm) and received 120 inescapable shocks (18 ~ 44s, average 30s; 0.45 mA for 15 s) once daily for 2 consecutive training days. After training, mice were subjected to 30 avoidance trials (18 ~ 44s, average 30s; 0.45 mA for 3s) 24 h after drug treatment. The inescapable times and latency were recorded.

### 2.3.5. Chronic mild stress (CMS) model

Rapid antidepressant potential were further tested for behavioral responses in mice exposed to CMS. The CMS protocol was designed to maximize the unpredictable nature of the stressors. CMS protocol followed as described previously CMS protocol with minor modifications (Schweizer et al., 2009). Mice were housed individually and received mild stress for 3 weeks. One of the following stressors was administered daily in an unpredictable random sequence over a three-week period: 24 h of food and water deficiency, 24 h of 45° cage tilt, 40 min of cage shaking (high speed horizontal shaking, 200 Transfer/min), the 20 h cage was dirty (200 ml of water in the sawdust pad), 12 h of nighttime illumination, and 50 ml of test tube for 6 h. All CMS mice were caged individually until the end of the experiment. Control mice were group housed without stress administration. One day post the last CMS procedure, mice received tests of a battery of behaviors in the order of NSF,

SPT, OFT, TST and FST, with at least 2 h resting time between tests. OFT, TST and FST were carried out on the second day post a single administration of the putatively effective fraction. NSF and SPT were tested on the following day morning.

### 2.3.6. Sucrose preference test (SPT)

The procedure for SPT followed a published procedure with minor modifications (Gross and Pinhasov, 2016). Briefly, mice were housed individually and exposed to sucrose solution (2% in tap water, Sigma, St Louis, Mo, USA) for 72 h, followed by 18 h of dehydration and 2 h exposure to two identical bottles. One is filled with 2% sucrose solution and the other is filled with water. The weight of the sucrose solution and water exposed for 2 h was measured. Sucrose preference was defined as the ratio of sucrose weight to total weight (sucrose + water) during the 2 h test, normalized to the body weight of each animal.

### 2.3.7. Novelty-suppressed feeding (NSF) test

In the NSF test, mice were deprived of food for 18 h and then placed in an open field (as described above). An ordinary rat chowder was placed in the center of the open air arena. Each animal is initially placed in a corner of the arena and allowed to explore for 10 min. The test ends in 10 min or when the mouse chews the mouse's food. Food consumption in the family cage is also measured as a control measure of appetite. The latency to begin eating, which is defined as the time until the mouse actively chewed the pellet, was recorded.

In the present study, we first screened the optimal dose of YG for rapid antidepressant potential by taking advantage of the fact that, up to now, all of the rapid antidepressant response was associated with a long-lasting antidepressant activity after a single dosing. Therefore, we screened the doses that induced persistent antidepressant response in TST or FST. By using this dose, we then tested the rapid antidepressant activity in mice subjected to chronic mild stress or learned helplessness (Autry et al., 2011; Li et al., 2010; Xue et al., 2013).

### 2.4. Western blot

The entire hippocampus (ventral and dorsal) was dissolved in RIPA buffer containing protease inhibitors and phosphatase inhibitors. Protein concentration was determined by BCA colorimetry (Pierce, Rockford, IL, USA). The protein lysate was separated by 12% SDS-PAGE electrophoresis and transferred to a polyvinylidene fluoride (PVDF) membrane. After blocking with 3% BSA for 1 h, NR1, NR2A, NR2B (1:1000, Cell Signaling Technology) and  $\beta$ -tubulin (1:5000, Proteintech Group, Rosemont, IL, USA) antibody were diluted with 1% BSA and incubated at room temperature for 6 h, then with horseradish. The oxidase was incubated with the secondary antibody for 2 h. Spots were then visualized using a super-signal Sipico chemiluminescent substrate (Thermo Fisher Scientific Inc.). NR1, NR2A and NR2B were normalized to the  $\beta$ -tubulin band. All experiments were performed at least 3 times.

### 2.5. Statistics analysis

The treatment effect from two samples was analyzed with student t-test, two-tailed, for two samples. One-way ANOVA with LSD comparison test for independent measurement, or Two-way ANOVA with LSD post hoc test for repeated measurement, was applied. All data are presented as Mean  $\pm$  SEM, statistical significance of 5% levels is acceptable.

## 3. Results

### 3.1. A single dose of YG elicited immediate and lasting antidepressant effects in TST and FST

Behavior despair paradigms TST and FST were used to first screen rapid antidepressant potential of YG. Twenty-four hours after a single

administration of YG, the immobility time was reduced significantly both in TST (Fig. 2A, ANOVA,  $F(5,47) = 14.36$ ,  $p < 0.001$ ) and FST (Fig. 2B, ANOVA,  $F(5,47) = 8.033$ ,  $p = 0.001$ ) only at the dose of 2.5 g/kg, corresponding to the clinical dose in human.

We then carried out a time course study of the antidepressant activity after a single 2.5 g/kg dose of YG. Mice were tested with TST at 30 min, 24 h, 5 d and 7 d after a single administration of YG, and FST was performed at 2 h after TST. The immobility time was reduced significantly from 30 min to 7 d in TST (Fig. 2C, ANOVA, 30 min:  $F(2,23) = 9.697$ ,  $p < 0.05$ ; 24 h:  $F(2,26) = 7.420$ ,  $p < 0.05$ ; 5 d:  $F(2,26) = 6.17$ ,  $p < 0.01$ , 7 d:  $F(2,29) = 6.302$ ,  $p < 0.05$ ). The similar effects were also observed in FST (Fig. 2D, ANOVA, 30 min:  $F(2,23) = 10.137$ ,  $p < 0.05$ ; 24 h:  $F(2,26) = 11.43$ ,  $p < 0.05$ ; 5 d:  $F(2,26) = 11.06$ ,  $p < 0.05$ ; 7 d:  $F(2,29) = 4.298$ ,  $p < 0.05$ ). YG showed no significant in total distance or time spent in the center (data not shown).

### 3.2. Rapid antidepressant-like effects of YG on mice exposed to CMS

Chronic mild stress (CMS) model was used to further examine the rapid antidepressant-like response. Mice were subjected to CMS for 21 days. The body weight of the stress group began to decrease on the 7th day compared to control group, while the sucrose preference started to decrease on the 14th day (Figs. S2A and S2B). Following the last stress exposure at the 21 day (Fig. 3A), mice received a single administration of YG. One day later, mice were tested with a battery of behavioral tests in NSF, SPT, OFT, TST and FST, with 2 h resting time in between. In the NSF test, CMS mice showed longer latency to eat and less food consumption than the control group and YG reversed the deficits from CMS, similar to ketamine (Fig. 3B, ANOVA, Latency to food:  $F(3,35) = 52.42$ ,  $p < 0.001$  and Fig. 3C, ANOVA, Food consumption:  $F(3,35) = 49.52$ ,  $p < 0.001$ ). CMS mice showed a significant decrease in sucrose preference, an measurement of anhedonia, compared to control group. A single treatment with YG significantly increased sucrose preference at 24 h (Fig. 3D, ANOVA,  $F(3,35) = 58.06$ ,  $p < 0.001$ ). Compared with the control group, the immobility time of CMS group was significantly increased both in TST and FST, reversed by YG or ketamine (Fig. 3E, ANOVA, TST:  $F(3,35) = 22$ ,  $p < 0.001$ ; Fig. 3F, FST,  $F(3,35) = 6.596$ ,  $p < 0.001$ ). In the OFT, there is no alteration of the CMS on the total distance traveled or time spent in the center area, indicating that the CMS procedure did not change the open field behavior. Both YG and ketamine showed no significant in total distance or time spent in the center (data not shown).

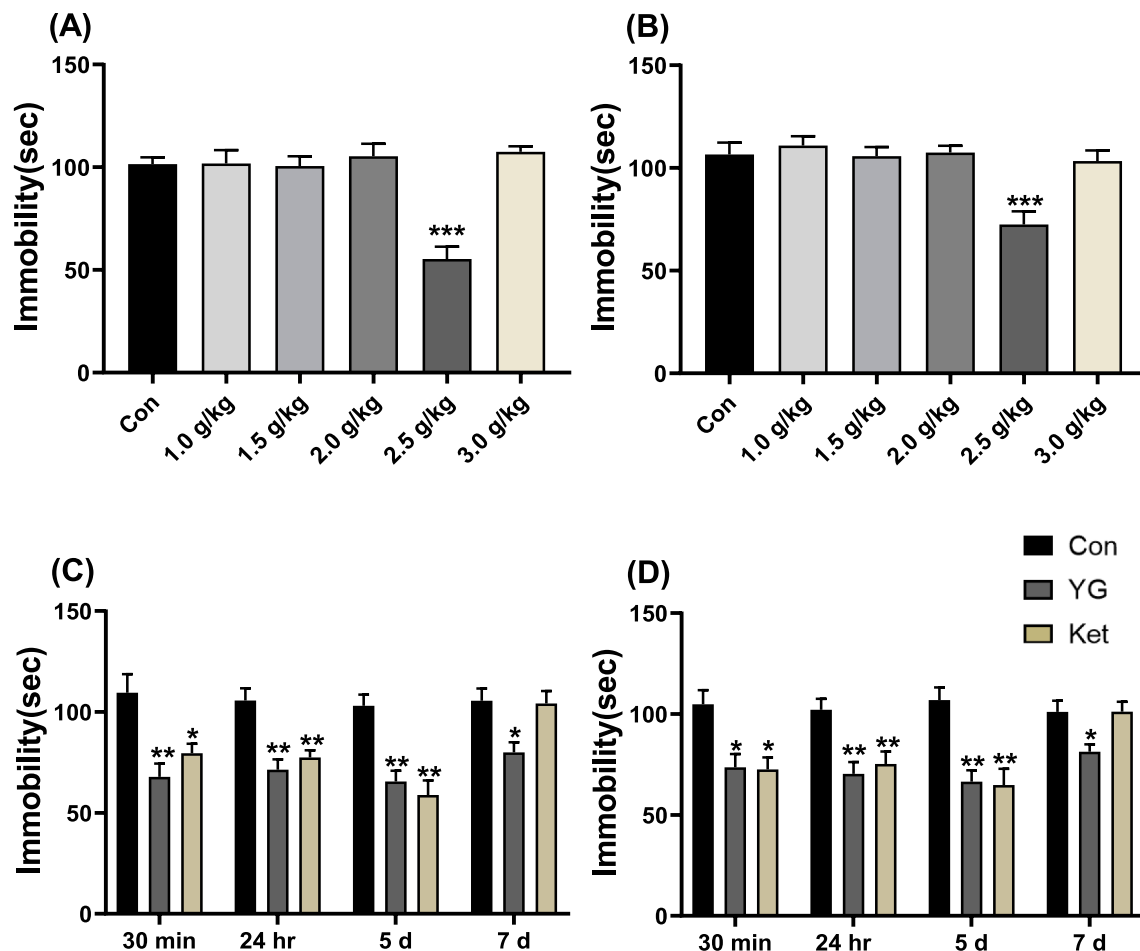
### 3.3. YG rapidly alleviated behavioral deficits in learned helplessness

Mice received inescapable shock training for two days, followed with a single dose of YG. Learned helplessness response was tested 24 h later. Compared with control group, the latency to escape and the number of escape failure were significantly increased in the LH mice. Similar to ketamine, YG treatment significantly reduced the latency to escape and the number of escape failures (Fig. 4A and B, ANOVA, Latency:  $F(3,31) = 9.576$ ,  $p = 0.0016$ ; Escape failure times:  $F(3,31) = 8.53$ ,  $p < 0.001$ ).

### 3.4. Time-course changes of the expression of NMDA receptor subunits in the hippocampus after a single administration of YG

After a single dose of YG administration, the expression of NR1 was significantly reduced from 30 min to 5 d, compared to control group, which was similar to ketamine (Fig. 5A, ANOVA, 30 min: ( $F(2,17) = 80.44$ ,  $p < 0.001$ ), 24 h:  $F(2,17) = 4.000$ ,  $p = 0.003$ ); 5 d:  $F(2,17) = 30.29$ ,  $p = 0.021$ ). Furthermore, the expression of NR2A and NR2B were both reduced in hippocampus 30 min after a single administration of YG, but these expressions returned normal at 24 h (Fig. 5B and C, ANOVA, NR2A: 30 min:  $F(2,17) = 1.16$ ,  $p = 0.047$ ,





**Fig. 2.** Screen of rapid antidepressant potential of YG. The effective dose at 24 h after a single administration of YG. (A and B) Immobility time was measured for the last 4 min during the 6 min testing time of TST and FST in groups of Con (control), 1.0 g/kg, 1.5 g/kg, 2.0 g/kg, 2.5 g/kg and 3.0 g/kg of YG. (C and D) Time course study of the antidepressant activity after a single administration of YG. Independent cohort of mice were used to test the behaviors at each time point. \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ , compared with control group, One Way ANOVA,  $n = 8-10$ /group.

24 h:  $F(2,17) = 2.864$ ,  $p = 0.232$ ; NR2B: 30 min:  $F(2,17) = 0.068$ ,  $p = 0.041$ , 24 h:  $F(2,17) = 3.533$ ,  $p = 0.200$ .

### 3.5. A single dose of YG normalized the upregulated expressions of NMDA receptor subunit NR1 in the hippocampus both in CMS and LH exposed mice

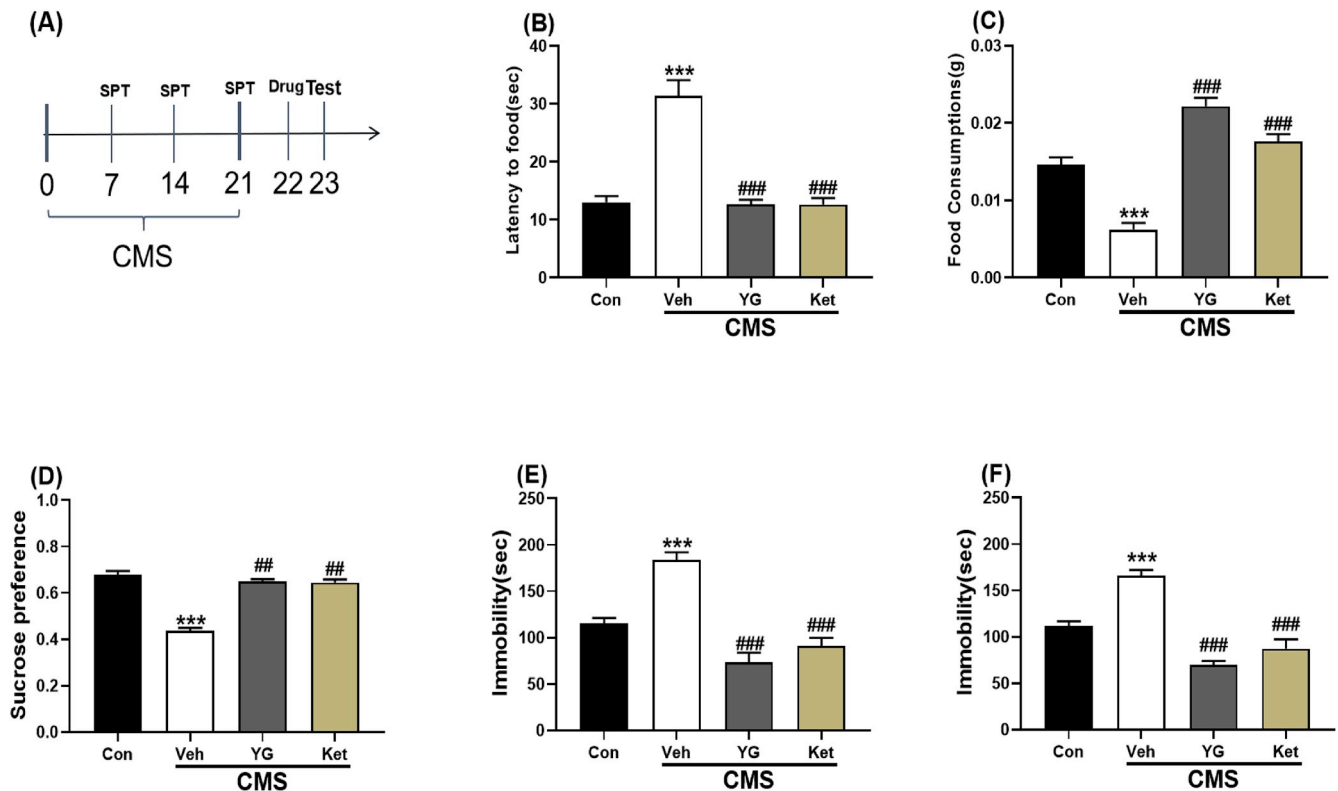
As only NR1 showed immediate and lasting change post a single dose of YG, We focused on evaluating the expression of NR1 both in LH and CMS mice after a single dose of YG. The expression of NR1 was up-regulated both in both models of LH and CMS (Fig. 6A and B, ANOVA, CMS:  $F(3,23) = 30.148$ ,  $p < 0.001$ ; LH:  $F(3,23) = 41.893$ ,  $p < 0.001$ ), compared to control group. Twenty-four hours after a single administration of YG, the expression of NR1 was attenuated, not different from the normal animals, similar to ketamine.

### 3.6. The impact of NMDA/NO/cGMP signaling on the antidepressant activity of YG

The impact of NMDA/NO/cGMP signaling on YG's antidepressant effects were examined in two conditions. First, we tested the synergy of the antagonists/inhibitors of the signaling with YG, by using combination of a subeffective dose of the antagonists/inhibitors with a sub-effective dose of YG. Second, we tested the prevention of the antidepressant response of YG by pretreatment of the agonists of the signaling.

#### 3.6.1. The impact of NMDA signaling

A subeffective dose of MK-801 (0.05 mg/kg), a NMDAR antagonist, was co-administered with a subeffective dose of YG (1 g/kg), which significantly reduced the immobility time, compared to control group, in TST (Fig. 7A. Two-way ANOVA, main effect of treatment,  $F(1, 38) = 25.17$ ,  $p < 0.001$ ; pre-treatment,  $F(1, 38) = 35.7$ ,  $p < 0.001$ ; treatment  $\times$  pre-treatment interaction,  $F(1, 38) = 47.71$ ,  $p < 0.001$ ), whereas administration of either one failed to result in behavioral changes. Similar findings were also detected in FST (Fig. 7B. Two-way ANOVA, main effect of treatment,  $F(1, 39) = 27.473$ ,  $p < 0.001$ ; pre-treatment,  $F(1, 39) = 25.092$ ,  $p < 0.001$ ; treatment  $\times$  pre-treatment interaction,  $F(1, 39) = 45.581$ ,  $p < 0.001$ ). Conversely, pretreatment of NMDA blunted the antidepressant effect of YG in TST or FST. NMDA (75 mg/kg, i.p.), when given alone, had no significant effect on the immobility time in TST or FST. However, the reduction of immobility time by YG (2.5 g/kg) was reversed when NMDA was pretreated, and the immobility time no longer differed from control group in TST (Fig. 7C. Two-way ANOVA, main effect of treatment,  $F(1, 38) = 76.111$ ,  $p < 0.001$ ; pre-treatment,  $F(1, 38) = 42.823$ ,  $p < 0.001$ ; treatment  $\times$  pre-treatment interaction,  $F(1, 38) = 74.193$ ,  $p < 0.001$ ) or FST (Fig. 7D. Two-way ANOVA, main effect of treatment,  $F(1, 39) = 14.486$ ,  $p = 0.001$ ; pre-treatment,  $F(1, 39) = 17.23$ ,  $p < 0.001$ ; treatment  $\times$  pre-treatment interaction,  $F(1, 39) = 10.672$ ,  $p = 0.002$ ). No significant effect was found in OFT for any treatment (data not shown).

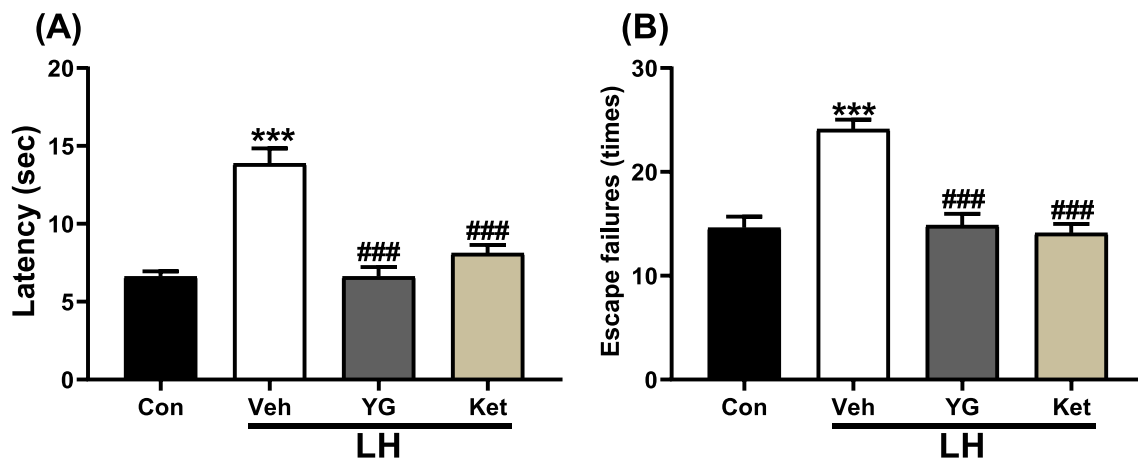


**Fig. 3.** Rapid antidepressant behaviors following a single dose of YG in mice exposed to chronic mild stress. (A) Mice were exposed to chronic mild stress (CMS), and received treatment of Veh (Vehicle), YG and ketamine (Ket). Control (Con) mice were not exposed to stress but received vehicle treatment. Behaviors were tested 1 day after drug administration. (B and C) Group difference in latency to food and food consumption in NSF test; (D) Group difference in sucrose preference test. (E and F) Group difference in immobility time in TST or FST <sup>\*\*\*</sup> $p < 0.001$ , compared with non-stress control group, <sup>###</sup> $p < 0.001$ , compared with vehicle group, One Way ANOVA,  $n = 9/\text{group}$ .

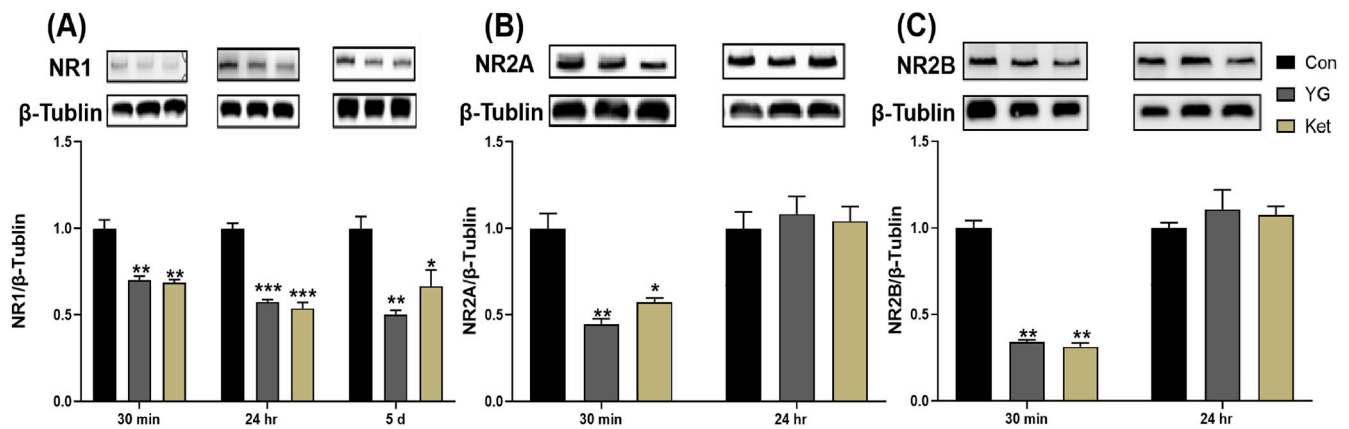
### 3.6.2. The impact of NO signaling

The role of NO on YG's antidepressant action was assessed by using a NO agonist L-Arginine or NO inhibitor L-NAME or 7-nitroindazole on the antidepressant effect of YG. Fig. 8A shows that L-Arginine (750 mg/kg, i.p.) inhibited the antidepressant-like effects of YG (2.5 g/kg) in TST (Two-way ANOVA, main effect of treatment,  $F(1, 39) = 22.359$ ,  $p < 0.001$ ; pre-treatment,  $F(1, 39) = 31.756$ ,  $p < 0.001$ ; treatment  $\times$  pre-treatment interaction,  $F(1, 39) = 28.635$ ,  $p < 0.001$ ). Fig. 8B illustrates that L-Arginine blunted the antidepressant-like effects of YG in FST (Two-way ANOVA, main effect of treatment,  $F(1,$

$39) = 80.952$ ,  $p < 0.001$ ; pre-treatment,  $F(1, 39) = 78.919$ ,  $p < 0.001$ ; treatment  $\times$  pre-treatment interaction,  $F(1, 39) = 96.709$ ,  $p < 0.001$ ). No effect was found in OFT (data not shown). As shown in Fig. 8C–D, co-administration of non-effective L-NAME (10 mg/kg, i.p.) and YG (1 g/kg, i.g.) had significant antidepressant-like activity in TST (Two-way ANOVA, main effect of treatment,  $F(1, 39) = 3.575$ ,  $p < 0.05$ ; pre-treatment,  $F(1, 39) = 18.404$ ,  $p < 0.001$ ; treatment  $\times$  pre-treatment interaction,  $F(1, 39) = 10.241$ ,  $p = 0.003$ ) and FST (Two-way ANOVA, main effect of treatment,  $F(1, 39) = 37.458$ ,  $p < 0.001$ ; pre-treatment,  $F(1, 39) = 33.903$ ,  $p < 0.001$ ; treatment  $\times$  pre-treatment interaction,  $F(1,$



**Fig. 4.** Rapid antidepressant effects of YG in the learned helplessness paradigm. Mice were exposed to learned helplessness (LH), and received treatment of Veh (vehicle), YG and ketamine (Ket). Control (Con) mice were not exposed to LH but received vehicle treatment. Behaviors were tested 1 day after drug administration. (A and B) Latency to escape and the number of escape failures in LH. <sup>\*</sup> $p < 0.05$  compared with non-stress control, <sup>#</sup> $p < 0.05$ , <sup>##</sup> $p < 0.05$  compared with vehicle group, One Way ANOVA,  $n = 8/\text{group}$ .



**Fig. 5.** NMDAR subunit expressions in the hippocampus at different times post a single administration of YG and ketamine. (A) NR1 protein expression levels at 30 min or 5 days after a single administration of YG. (B, C) NR2A and NR2B protein expression levels at 30 min after YG. \*\* $p < 0.01$ , \* $p < 0.05$  compare with control group, One Way ANOVA,  $n = 6/\text{group}$ .

(1, 39) = 32.858,  $p < 0.001$ ). Fig. 8E demonstrates the effects of a specific neuronal NOS inhibitor; 7-nitroindazole (30 mg/kg, i.p.) in combination with YG (1 g/kg, i.g.) on the antidepressant-like effect in TST when each drug was administered alone or in combination (Two-way ANOVA, main effect of treatment,  $F(1, 28) = 21.081$ ,  $p < 0.001$ ; pre-treatment,  $F(1, 28) = 26.7214$ ,  $p < 0.001$ ; treatment  $\times$  pre-treatment interaction,  $F(1, 28) = 17.343$ ,  $p < 0.001$ ). Fig. 8F shows the similar effect in FST (Two-way ANOVA, main effect of treatment,  $F(1, 28) = 45.893$ ,  $p < 0.001$ ; pre-treatment,  $F(1, 28) = 79.840$ ,  $p < 0.001$ ; treatment  $\times$  pre-treatment interaction,  $F(1, 28) = 50.023$ ,  $p < 0.001$ ). No significant change was found in OFT (data not shown).

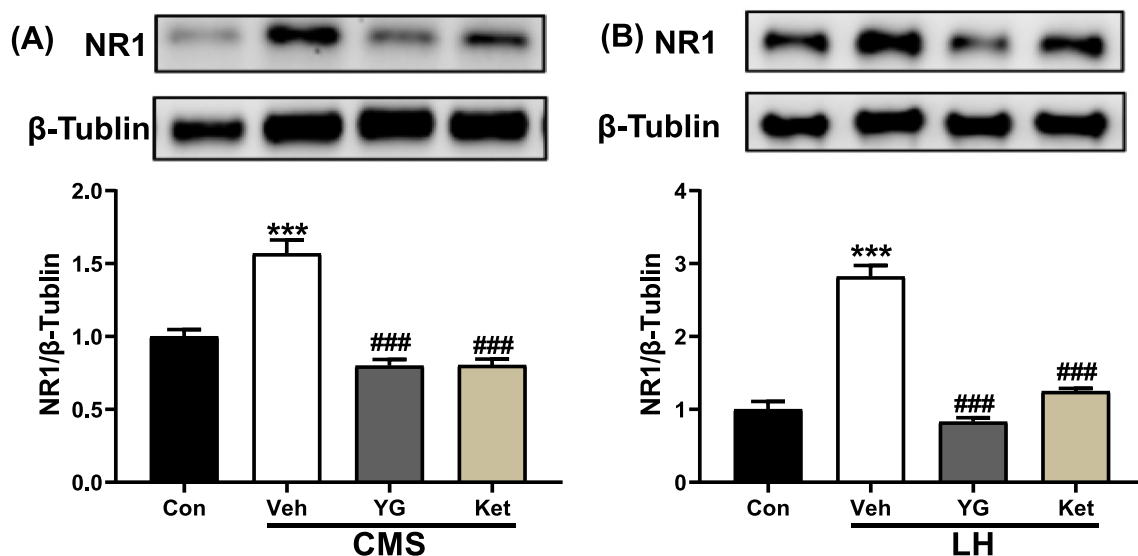
### 3.6.3. The impact of cGMP

As shown in Fig. 9A, Methylene blue (10 mg/kg, i.p.), a direct inhibitor of both NOS and sGC, was unable to affect the immobility time either TST or FST when used alone. However, when it was co-administered with a subeffective dose of YG (1 g/kg, i.g.), it considerably increased the antidepressant like activity both in TST (Two-way ANOVA, main effect of treatment,  $F(1, 39) = 22.305$ ,  $p < 0.001$ ; pre-treatment,  $F(1, 39) = 18.205$ ,  $p < 0.001$ ; treatment  $\times$  pre-treatment interaction,  $F(1, 39) = 19.176$ ,  $p < 0.001$ ) and FST (Fig. 9B; Two-way ANOVA, main effect of treatment,  $F(1, 39) = 39.814$ ,  $p < 0.001$ ; pre-treatment,  $F(1, 39) = 36.056$ ,  $p < 0.001$ ; treatment  $\times$  pre-treatment

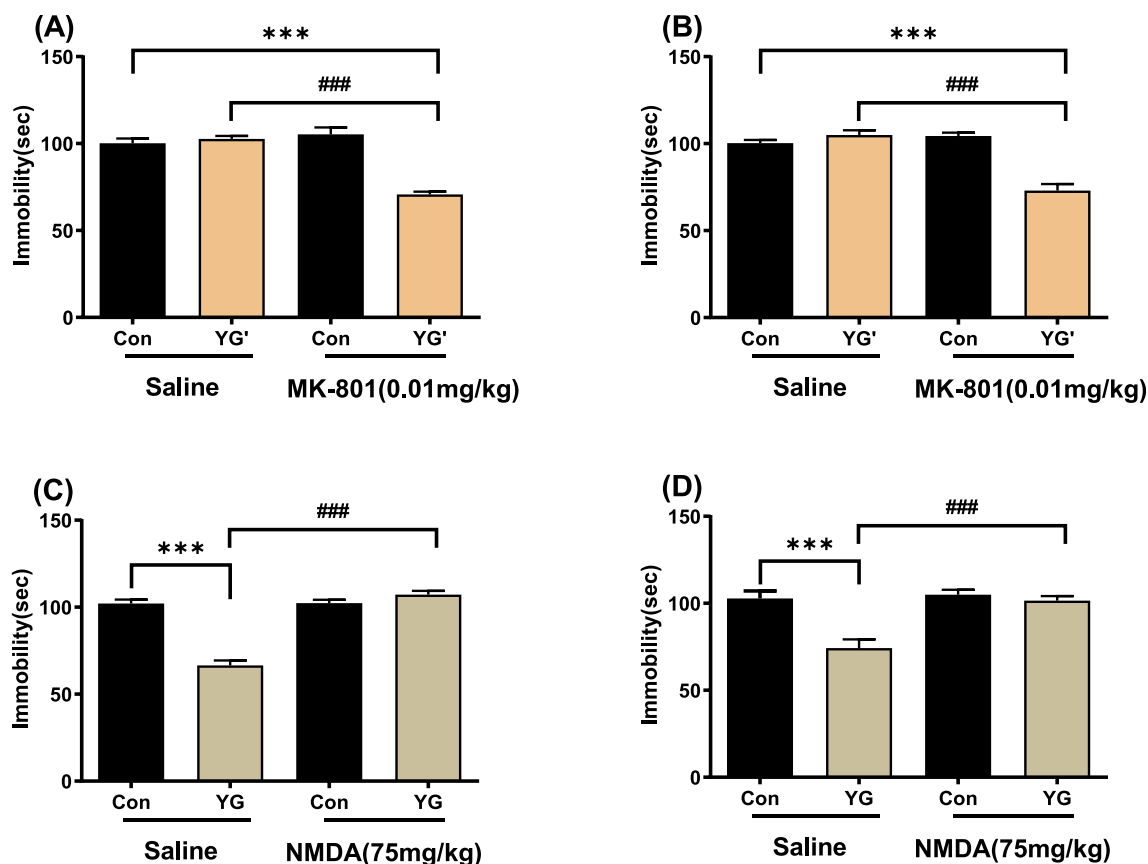
interaction,  $F(1, 39) = 34.174$ ,  $p < 0.001$ ). Administration of methylene blue alone or in combination with YG did not show a significant effect ( $p > 0.05$ , data not shown) in OFT. Fig. 9C and D showed the effect of sildenafil (5 mg/kg, i.p., a PDE5 inhibitor that increase the level of cGMP, after co-administrated with effective dose of YG in mice. Pretreatment with sildenafil alone failed to affect the immobility time of TST. However, when the mice were pretreated with sildenafil, the antidepressant effect of YG was blocked in TST (Fig. 9C; Two-way ANOVA, main effect of treatment,  $F(1, 31) = 16.828$ ,  $p < 0.001$ ; pre-treatment,  $F(1, 31) = 32.030$ ,  $p < 0.001$ ; treatment  $\times$  pre-treatment interaction,  $F(1, 31) = 20.901$ ,  $p < 0.001$ ) and FST (Fig. 9D; Two-way ANOVA, treatment,  $F(1, 39) = 34.282$ ,  $p < 0.001$ ; pre-treatment,  $F(1, 39) = 31.258$ ,  $p < 0.05$ ; treatment  $\times$  pre-treatment interaction,  $F(1, 39) = 33.729$ ,  $p < 0.001$ ). Sildenafil administration, alone or in combination with YG, did not show a significant effect in OFT (data not shown).

## 4. Discussion

The present study investigated the potential of YG for fast-onset antidepressant therapy. We found a single dose of YG elicited antidepressant effects at 30 min, and lasted for 7 days. We also found a single dose of YG was capable to reverse the depression-like behaviors



**Fig. 6.** NR1 expressions in the hippocampus post a single administration of YG and ketamine at 24 h in CMS and LH. \*\* $P < 0.01$ , compared with control group; ### $p < 0.001$ , compared with CMS or LH model group, One Way ANOVA,  $n = 6/\text{group}$ .



**Fig. 7.** Antidepressant effects of YG following NMDA signaling intervention. Effects of NMDA antagonist or NMDA pretreatment on antidepressant response of YG (1 g/kg, i.g.) in TST (A) and FST (B). Effects of NMDA (75 mg/kg, i.p.) pretreatment on antidepressant response of YG (2.5 g/kg, i.g.) in the TST (C) and FST (D). \*\*\* $p < 0.001$  compared with Control + Saline; ### $p < 0.001$  compared with Saline + YG or YG'. Two Way ANOVA,  $n = 9-10$ /group.

in chronically stressed or learned helpless mice, and thus confirmed the rapid antidepressant-like effect. Furthermore, a single dose of YG reversed the up-regulated expressions of NMDA receptor subunit, particularly NR1, in the hippocampus of CMS and LH mice. The role of NMDA/NO/cGMP in the antidepressant action of YG was systematically tested using various pharmacological interventions. We found that the antidepressant-like effect of YG was blunted by the pretreatment with the agonists of the signaling including NMDA, L-Arginine or sildenafil. Conversely, administration of the antagonists of the signaling including MK-801, 7-nitroindazole, methylene blue facilitated the antidepressant effect of a subeffective dose of YG. These findings suggested that YG elicited rapid antidepressant responses, and its inhibition of NMDA/NO/cGMP signaling played an important role.

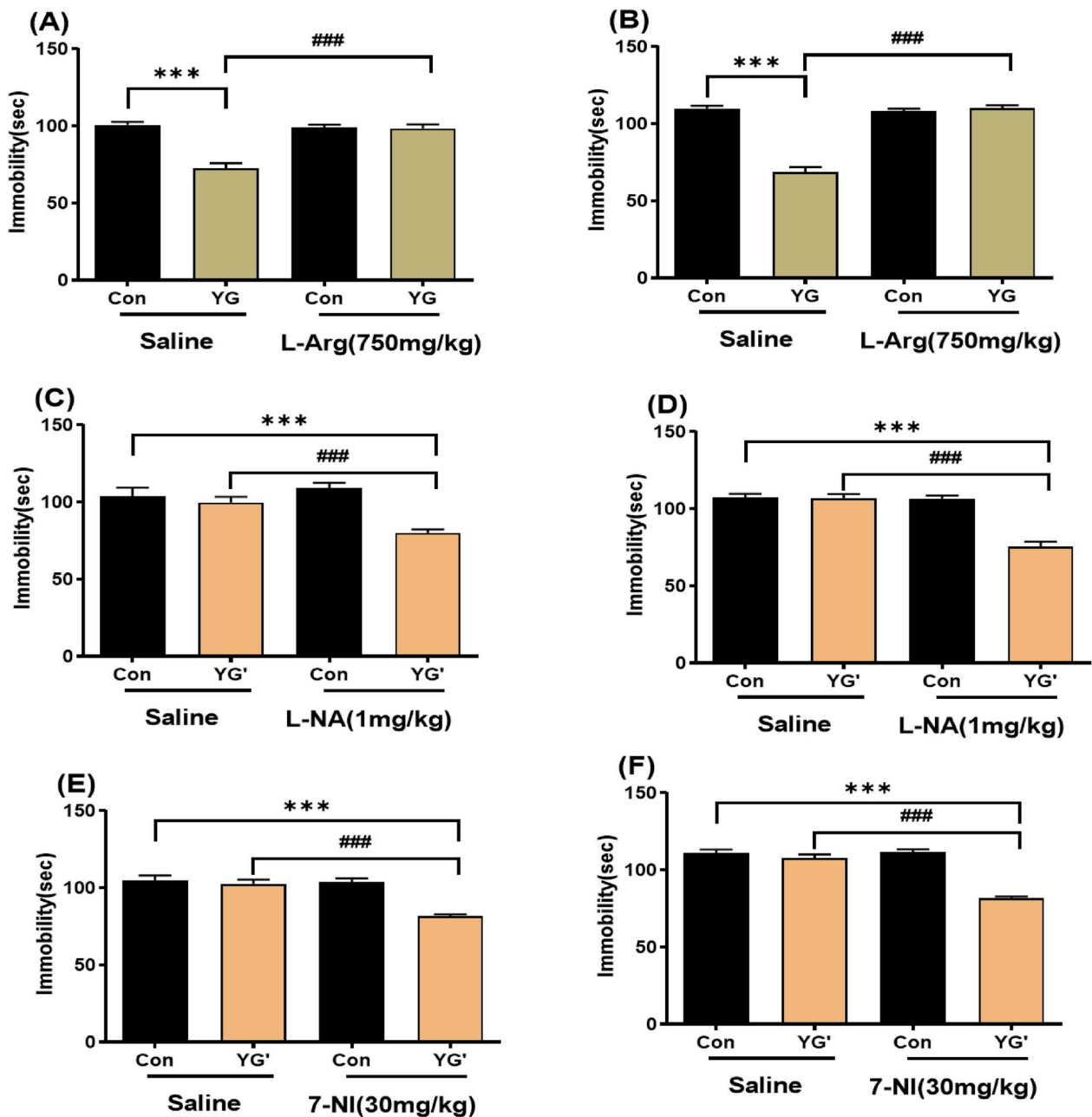
Several evidence indicates the rapid antidepressant potential of YG. First, a single dose of YG elicited immediate and 7 days long antidepressant activities in behavior despair paradigms. The long antidepressant activity measured in these behavior has been reported for ketamine or Yueju (Tang et al., 2015; Xue et al., 2013, 2016). The long-lasting antidepressant effect of YG or ketamine may reflect the long-term improvement of neural plasticity, which is in contrast to the short-lived antidepressant activity by a single dose of SSRI. More importantly the rapid antidepressant-like activity of YG was evidenced in the animal models of depression including CMS and LH, in which an antidepressant response requires repeated administration of SSRIs (Duman, 2018; Duman et al., 2012; Li et al., 2010). Following CMS, mice showed a full spectrum of depression-like and/or anxiety-like symptoms, including anhedonia response, increased immobility time in TST and FST (Gross and Pinhasov, 2016). Similar to ketamine, the depression-like behaviors in TST, FST, SPT, and NSF in mice exposed to CMS were reversed by only a single dose of YG (Autry et al., 2011). Similarly, the

helpless responses in LH with increased escape failures and latency to escape were also attenuated by a single dose of YG (Maeng et al., 2008; Mingmalairak et al., 2010). These converging findings supported the rapid antidepressant effects of YG.

The present study focused on NMDA signaling in antidepressant activity. Some studies have demonstrated that in the hippocampus, a single dose of drug such as Yueju or ketamine reduced expressions of NMDA receptors including NR1, NR2A, and NR2B (Xia et al., 2016). We found that YG induced an immediate down-regulation of NR1, NR2A and NR2B, but only NR1 was lastingly down-regulated. This was similar to the effects of Yueju pill alone, except that NR2A was not changed (Xia et al., 2016). Therefore, these findings support that the immediately and lastingly reduced NR1 expression is temporally linked to the antidepressant effects, whereas YG has stronger initial inhibitory action than Yueju by additional suppression of both NR2A and NR2B expression. Furthermore, YG also corrected the abnormality of NMDA signaling in the hippocampus of LH and CMS mice: The expression of NR1 was increased in both CMS and LH. After a single YG administration, NR1 expression was reversed, similar to Yueju or ketamine (Tang et al., 2015; Xia et al., 2016). The data suggested that the attenuation of NR1 expression, and hence, of NMDA activity, could be part of the mechanism underlying the immediate and persistent antidepressant effects of YG, whereas enhancement of NMDA signaling in the hippocampus after chronic stress is pathophysiological to depression.

Increasing number of studies show that NMDA's downstream NO-cGMP pathway are involved in the pathogenesis of depression, such as ketamine (Araujo et al., 2018; Cunha et al., 2015; Haj-Mirzaian et al., 2016a,b; Khan et al., 2016) and antidepressant activity. In the present study, we systematically examined the effects of NMDA/NO/cGMP



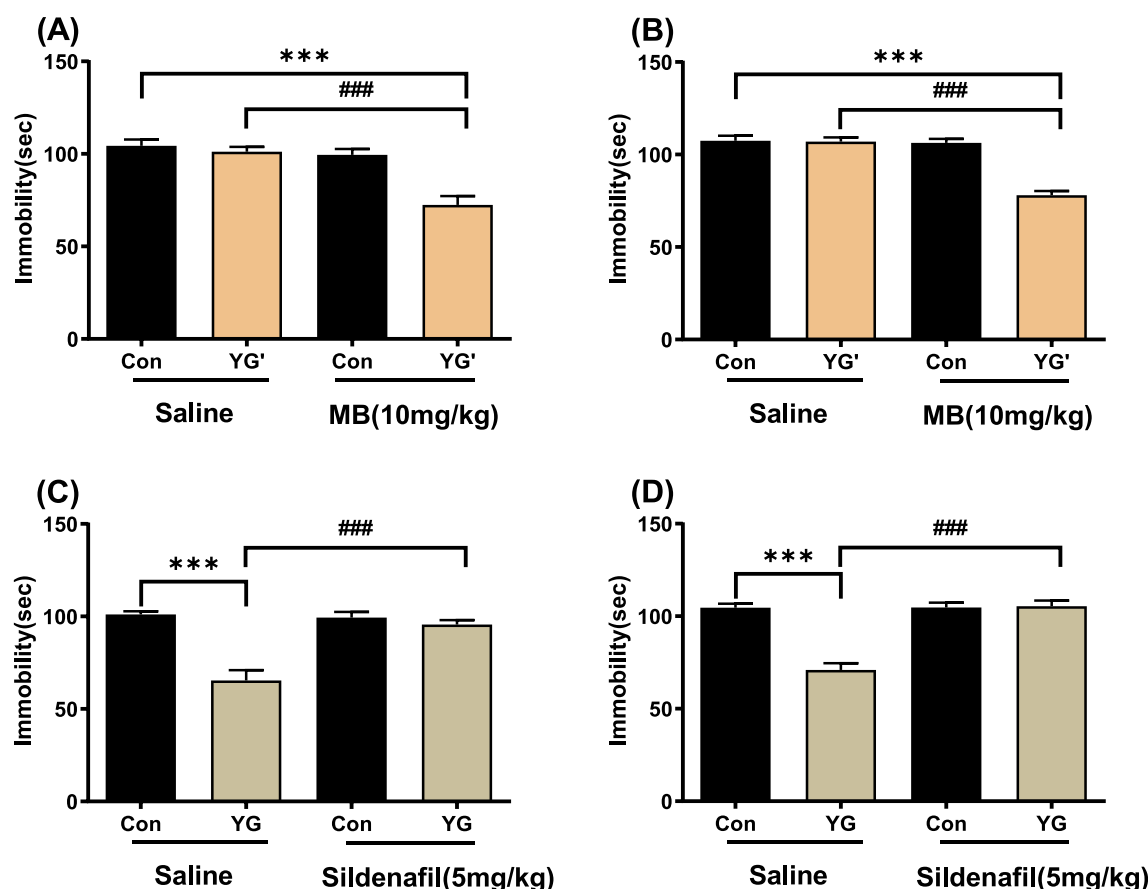


**Fig. 8.** Antidepressant effects of YG following NO signaling intervention. (A,B) L-Arg (750 mg/kg, i.p.) pretreatment on Antidepressant response of YG (2.5 g/kg, i.g) in TST and FST. The effect of L-NAME (10 mg/kg, i.p.) pretreatment on subeffective dose of YG (1 g/kg, i.g) in the TST (C) and FST (D). The effect of 7-NI (30 mg/kg, i.p.) pretreatment on antidepressant response of YG' (1 g/kg, i.g.) in the TST (E) and FST (F). \*\*\* $p < 0.001$ , compared with Control + Saline; ### $p < 0.001$ , compared with Saline + YG or YG'. Two Way ANOVA,  $n = 8-10$ /group.

signaling on the antidepressant effects of YG, and found that the effects were blunted by pretreatment of NMDA, but facilitated by NMDA antagonist (Araujo et al., 2018; Haj-Mirzaian et al., 2016a,b; Khan et al., 2016). Consistently, the pretreatment of agents to increase level of NO or cGMP blunted the antidepressant response of YG, whereas, drugs which inhibit the generation of NO and/or cGMP signaling induced antidepressant effects from a subeffective dose of YG. An in vitro study showed that inhibition of NMDA/NO/cGMP is common across different conventional antidepressants, and antidepressant response from ketamine and other neurotransmitters also involved inhibition of NMDA/NO/cGMP (Cunha et al., 2015; Li et al., 2006). The inhibitors of the signaling augment the antidepressant effects of SSRIs, whereas agonists of the signaling suppressed the antidepressant effects of fluoxetine (Haj-Mirzaian et al., 2016a,b; Harkin et al., 2004). Therefore, the

antidepressant effect of YG, similar to conventional antidepressants, requires the inhibition of NMDA/NO/cGMP signaling (Gaur and Kumar, 2010). It appears that this pathway is shared for antidepressant activity.

Taken together, our results demonstrate that YG has rapid antidepressant-like effects, lasting for a long time, in which attenuation of NMDA and the downstream NO-cGMP signaling play an important role. However, there are some limitations of the study. For example, the expressions of specific NOS in hippocampus which regulate NO generation remain to be determined. The spacial and temporal profile of the signaling has yet to know. The compounds critical for the effects of YG need to be identified. Additionally, although the doses of herbs in YG are within the range of clinical dose and side effects have not been noticed, the assessment of the toxicity of the dose remains to be



**Fig. 9.** Antidepressant effects of YG following cGMP signaling intervention. (A, B) Antidepressant effect of MB (10 mg/kg, i.p.) pretreatment followed by subeffective dose of YG (1 g/kg, i.g.) in the TST and FST. (C, D) Pretreatment of Sildenafil (5 mg/kg, i.p.) on antidepressant effect of YG (2.5 g/kg, i.g.) in the TST. \*\*\* $p < 0.001$ , compared with Control + Saline; ### $p < 0.001$ , compared with Saline + YG or YG'. Two Way ANOVA,  $n = 8-10$ /group.

determined. Finally, the effective compounds or ingredients of YG for rapid antidepressant activity remained unclear in the present study. Genepinside is most enriched in YG, and repeated administration of genepinside has been shown antidepressant activities (Cai et al., 2015; Wang et al., 2016; Zhao et al., 2018). Our previous study however demonstrated that genepinside alone was unable to induce rapid antidepressant response (Zhang et al., 2015). Therefore, the chemicals in YG responsible for rapid antidepressant activity remain to be determined.

Our findings supported the efficacy of a single dose of YG in quickly and lastingly relieving depressive symptoms. Our systematic analysis indicated that YG strongly inhibited NMDA subunit expression and the NMDA/NO/cGMP signaling played a key role in the antidepressant response of YG. These laid a basis for further clarifying the key underlying substrates and discovery of natural compounds for improvement of depression therapy.

#### Funding information

National Science Foundation of China (81673625, 81874374, 81603497, 81803478). Priority Academic Program Development of Jiangsu Higher Education Institutions (PAPD), and Integration of Chinese and Western Medicine.

#### Author contributions

G. C. and H. Z. designed the project. H. Z., Y. S., C. N., Y. Z. and R. G. performed behavioral tests. Y. Y., S. Q., X. G., L. S. and Q. S. performed Western Blot. W. X., W. W. and T. Z. performed drug preparation, analysis and quality control. G. C. and H. Z. wrote the manuscript. All

authors discussed and commented on the manuscript.

#### Declaration of competing interest

All authors declare that he/she has no conflict of interest.

#### Acknowledgement

This study was supported by the National Science Foundation of China (81673625, 81874374, 81603497, 81803478). Priority Academic Program Development of Jiangsu Higher Education Institutions (PAPD), and Integration of Chinese and Western Medicine.

#### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2019.112380>.

#### References

- Amiri, S., Alijanpour, S., Tirgar, F., Haj-Mirzaian, A., Amini-Khoei, H., Rahimi-Balaei, M., Rastegar, M., Ghaderi, M., Ghazi-Khansari, M., Zarrindast, M.R., 2016. NMDA receptors are involved in the antidepressant-like effects of capsaicin following amphetamine withdrawal in male mice. *Neuroscience* 329, 122–133.
- Araujo, J., Junior, J., Damasceno, M., Santos, S., Vieira-Neto, A.E., Lobo, M., Campos, A.R., Moreira, R.A., Monteiro-Moreira, A., 2018. Neuropharmacological characterization of frutalin in mice: evidence of an antidepressant-like effect mediated by the NMDA receptor/NO/cGMP pathway. *Int. J. Biol. Macromol.* 112, 548–554.
- Autry, A.E., Adachi, M., Nosyreva, E., Na, E.S., Los, M.F., Cheng, P.F., Kavalali, E.T., Monteggia, L.M., 2011. NMDA receptor blockade at rest triggers rapid behavioural antidepressant responses. *Nature* 475 (7354), 91–95.
- Bonnet, U., 2015. Long-term ketamine self-injections in major depressive disorder: focus

- on tolerance in ketamine's antidepressant response and the development of ketamine addiction. *J. Psychoact. Drugs* 47 (4), 276–285.
- Burgdorf, J., Zhang, X.L., Nicholson, K.L., Balster, R.L., Leander, J.D., Stanton, P.K., Gross, A.L., Kroes, R.A., Moskal, J.R., 2013. GLYX-13, a NMDA receptor glycine-site functional partial agonist, induces antidepressant-like effects without ketamine-like side effects. *Neuropsychopharmacology* 38 (5), 729–742.
- Cai, L., Li, R., Tang, W.J., Meng, G., Hu, X.Y., Wu, T.N., 2015. Antidepressant-like effect of geniposide on chronic unpredictable mild stress-induced depressive rats by regulating the hypothalamus-pituitary-adrenal axis. *Eur. Neuropsychopharmacol.* 25 (8), 1332–1341.
- Chen, G., Guo, X., 2017. Neurobiology of Chinese herbal medicine on major depressive disorder. *Int. Rev. Neurobiol.* 135, 77–95.
- Coumans, J.V., Palanisamy, S.K., McFarlane, J., Moens, P.D., 2016. Proteomic and microscopic strategies towards the analysis of the cytoskeletal networks in major neuropsychiatric disorders. *Int. J. Mol. Sci.* 17 (4).
- Cunha, M.P., Pazini, F.L., Ludka, F.K., Rosa, J.M., Oliveira, A., Budni, J., Ramos-Hryb, A.B., Lieberknecht, V., Bettio, L.E., Martin-de-Saavedra, M.D., Lopez, M.G., Tasca, C.L., Rodrigues, A.L., 2015. The modulation of NMDA receptors and L-arginine/nitric oxide pathway is implicated in the anti-immobility effect of creatine in the tail suspension test. *Amino Acids* 47 (4), 795–811.
- Donello, J.E., Banerjee, P., Li, Y.X., Guo, Y.X., Yoshitake, T., Zhang, X.L., Miry, O., Kehr, J., Stanton, P.K., Gross, A.L., Burgdorf, J.S., Kroes, R.A., Moskal, J.R., 2018. Positive N-Methyl-D-Aspartate receptor modulation by rapastinel promotes rapid and sustained antidepressant-like effects. *Int. J. Neuropsychopharmacol.*
- Du, J., Machado-Vieira, R., Maeng, S., Martinowich, K., Manji, H.K., Zarate, C.J., 2006. Enhancing AMPA to NMDA throughput as a convergent mechanism for antidepressant action. *Drug Discov. Today Ther. Strat.* 3 (4), 519–526.
- Duman, R.S., Li, N., Liu, R.J., Duric, V., Aghajanian, G., 2012. Signaling pathways underlying the rapid antidepressant actions of ketamine. *Neuropharmacology* 62 (1), 35–41.
- Duman, R.S., 2018. Ketamine and Rapid-Acting Antidepressants: a New Era in the Battle against Depression and Suicide F1000Res. pp. 7.
- Fan, G., Huang, H., Lin, Y., Zheng, G., Tang, X., Fu, Y., Wei, H., Zhao, L., Liu, Z., Wang, M., Wang, S., Li, Q., Fang, Z., Zhou, Y., Dai, F., Qiu, X., 2018. Herbal medicine foot bath for the treatment of diabetic peripheral neuropathy: protocol for a randomized, double-blind and controlled trial. *Trials* 19 (1), 483.
- Feng, D.D., Tang, T., Lin, X.P., Yang, Z.Y., Yang, S., Xia, Z.A., Wang, Y., Zheng, P., Wang, Y., Zhang, C.H., 2016. Nine traditional Chinese herbal formulas for the treatment of depression: an ethnopharmacology, phytochemistry, and pharmacology review. *Neuropsychiatric Dis. Treat.* 12, 2387–2402.
- Gaur, V., Kumar, A., 2010. Protective effect of desipramine, venlafaxine and trazodone against experimental animal model of transient global ischemia: possible involvement of NO-cGMP pathway. *Brain Res.* 1353, 204–212.
- Gross, M., Pinhasov, A., 2016. Chronic mild stress in submissive mice: marked polydipsia and social avoidance without hedonic deficit in the sucrose preference test. *Behav. Brain Res.* 298 (Pt B), 25–34.
- Haj-Mirzaian, A., Kordjazy, N., Amiri, S., Haj-Mirzaian, A., Amini-Khoei, H., Ostadhadhi, S., Dehpour, A., 2016a. Involvement of nitric oxide-cyclic guanosine monophosphate pathway in the antidepressant-like effect of tropisetron and ondansetron in mice forced swimming test and tail suspension test. *Eur. J. Pharmacol.* 780, 71–81.
- Haj-Mirzaian, A., Kordjazy, N., Ostadhadhi, S., Amiri, S., Haj-Mirzaian, A., Dehpour, A., 2016b. Fluoxetine reverses the behavioral despair induced by neurogenic stress in mice: role of N-methyl-D-aspartate and opioid receptors. *Can. J. Physiol. Pharmacol.* 94 (6), 599–612.
- Harkin, A., Connor, T.J., Burns, M.P., Kelly, J.P., 2004. Nitric oxide synthase inhibitors augment the effects of serotonin re-uptake inhibitors in the forced swimming test. *Eur. Neuropsychopharmacol.* 14 (4), 274–281.
- Hartell, N.A., 1996. Inhibition of cGMP breakdown promotes the induction of cerebellar long-term depression. *J. Neurosci.* 16 (9), 2881–2890.
- Hillhouse, T.M., Porter, J.H., 2014. Ketamine, but not MK-801, produces antidepressant-like effects in rats responding on a differential-reinforcement-of-low-rate operant schedule. *Behav. Pharmacol.* 25 (1), 80–91.
- Khan, M.I., Ostadhadhi, S., Zolfaghari, S., Etemaei, M.S., Hassanzadeh, G., Dehpour, A.R., 2016. The involvement of NMDA receptor/NO/cGMP pathway in the antidepressant like effects of baclofen in mouse force swimming test. *Neurosci. Lett.* 612, 52–61.
- Li, N., Lee, B., Liu, R.J., Banasr, M., Dwyer, J.M., Iwata, M., Li, X.Y., Aghajanian, G., Duman, R.S., 2010. mTOR-dependent synapse formation underlies the rapid antidepressant effects of NMDA antagonists. *Science* 329 (5994), 959–964.
- Li, Y.F., Zhang, Y.Z., Liu, Y.Q., Wang, H.L., Cao, J.B., Guan, T.T., Luo, Z.P., 2006. Inhibition of N-methyl-D-aspartate receptor function appears to be one of the common actions for antidepressants. *J. Psychopharmacol.* 20 (5), 629–635.
- Lou, J.S., Li, C.Y., Yang, X.C., Fang, J., Yang, Y.X., Guo, J.Y., 2010. Protective effect of gan mai da zao decoction in unpredictable chronic mild stress-induced behavioral and biochemical alterations. *Pharm. Biol.* 48 (12), 1328–1336.
- Maeng, S., Zarate, C.J., Du, J., Schloesser, R.J., McCammon, J., Chen, G., Manji, H.K., 2008. Cellular mechanisms underlying the antidepressant effects of ketamine: role of alpha-amino-3-hydroxy-5-methylisoxazole-4-propionic acid receptors. *Biol. Psychiatry* 63 (4), 349–352.
- Mingmalairak, S., Tohda, M., Murakami, Y., Matsumoto, K., 2010. Possible involvement of signal transducers and activators of transcription 3 system on depression in the model mice brain. *Biol. Pharm. Bull.* 33 (4), 636–640.
- Neitz, A., Mergia, B., Imbrosci, B., Petrasch-Parwez, E., Eysel, U.T., Koesling, D., Mittmann, T., 2014. Postsynaptic NO/cGMP increases NMDA receptor currents via hyperpolarization-activated cyclic nucleotide-gated channels in the hippocampus. *Cerebr. Cortex* 24 (7), 1923–1936.
- Read, S.J., Hirst, W.D., Upton, N., Parsons, A.A., 2001. Cortical spreading depression produces increased cGMP levels in cortex and brain stem that is inhibited by tonabersat (SB-220453) but not sumatriptan. *Brain Res.* 891 (1–2), 69–77.
- Schweizer, M.C., Henniger, M.S., Sillaber, I., 2009. Chronic mild stress (CMS) in mice: of anhedonia, 'anomalous anxiolysis' and activity. *PLoS One* 4 (1), e4326.
- Tang, J., Xue, W., Xia, B., Ren, L., Tao, W., Chen, C., Zhang, H., Wu, R., Wang, Q., Wu, H., Duan, J., Chen, G., 2015. Involvement of normalized NMDA receptor and mTOR-related signaling in rapid antidepressant effects of Yueju and ketamine on chronically stressed mice. *Sci. Rep.* 5, 13573.
- Tanti, A., Belzung, C., 2010. Open questions in current models of antidepressant action. *Br. J. Pharmacol.* 159 (6), 1187–1200.
- Thierry, B., Steru, L., Chermat, R., Simon, P., 1984. Searching-waiting strategy: a candidate for an evolutionary model of depression? *Behav. Neural. Biol.* 41 (2), 180–189.
- Vuckovic, S., Srebro, D., Savic, V.K., Prostran, M., 2015. The antinociceptive effects of magnesium sulfate and MK-801 in visceral inflammatory pain model: the role of NO/cGMP/K(+)ATP pathway. *Pharm. Biol.* 53 (11), 1621–1627.
- Wang, J., Duan, P., Cui, Y., Li, Q., Shi, Y., 2016. Geniposide alleviates depression-like behavior via enhancing BDNF expression in hippocampus of streptozotocin-evoked mice. *Metab. Brain Dis.* 31 (5), 1113–1122.
- Wei, X.H., Cheng, X.M., Shen, J.S., Wang, Z.T., 2008. Antidepressant effect of Yueju-Wan ethanol extract and its fractions in mice models of despair. *J. Ethnopharmacol.* 117 (2), 339–344.
- Wu, R., Zhu, D., Xia, Y., Wang, H., Tao, W., Xue, W., Xia, B., Ren, L., Zhou, X., Li, G., Chen, G., 2015. A role of Yueju in fast-onset antidepressant action on major depressive disorder and serum BDNF expression: a randomly double-blind, fluoxetine-adjunct, placebo-controlled, pilot clinical study. *Neuropsychiatric Dis. Treat.* 11, 2013–2021.
- Xia, B., Zhang, H., Xue, W., Tao, W., Chen, C., Wu, R., Ren, L., Tang, J., Wu, H., Cai, B., Doron, R., Chen, G., 2016. Instant and lasting down-regulation of NR1 expression in the Hippocampus is associated temporally with antidepressant activity after acute Yueju. *Cell. Mol. Neurobiol.* 36 (7), 1189–1196.
- Xue, W., Zhou, X., Yi, N., Jiang, L., Tao, W., Wu, R., Wang, D., Jiang, J., Ge, X., Wang, Y., Wu, H., Chen, G., 2013. Yueju pill rapidly induces antidepressant-like effects and acutely enhances BDNF expression in mouse brain. *Evid Based Complement Alternat Med* 2013, 184367.
- Xue, W., Wang, W., Gong, T., Zhang, H., Tao, W., Xue, L., Sun, Y., Wang, F., Chen, G., 2016. PKA-CREB-BDNF signaling regulated long lasting antidepressant activities of Yueju but not ketamine. *Sci. Rep.* 6, 26331.
- Yeung, W.F., Chung, K.F., Ng, K.Y., Yu, Y.M., Ziea, E.T., Ng, B.F., 2014. A meta-analysis of the efficacy and safety of traditional Chinese medicine formula Ganmai Dazao decoction for depression. *J. Ethnopharmacol.* 153 (2), 309–317.
- Zhang, H., Xue, W., Wu, R., Gong, T., Tao, W., Zhou, X., Jiang, J., Zhang, Y., Zhang, N., Cui, Y., Chen, C., Chen, G., 2015. Rapid antidepressant activity of ethanol extract of *Gardenia jasminoides* Ellis is associated with upregulation of BDNF expression in the Hippocampus. *Evid. Based Complement Altern. Med.* 2015, 761238.
- Zhao, Y., Li, H., Fang, F., Qin, T., Xiao, W., Wang, Z., Ma, S., 2018. Geniposide improves repeated restraint stress-induced depression-like behavior in mice by ameliorating neuronal apoptosis via regulating GLP-1R/AKT signaling pathway. *Neurosci. Lett.* 676, 19–26.
- Zhao, Z., Wang, W., Guo, H., Zhou, D., 2008. Antidepressant-like effect of liquiritin from *Glycyrrhiza uralensis* in chronic variable stress induced depression model rats. *Behav. Brain Res.* 194 (1), 108–113.
- Zomkowski, A.D., Engel, D., Cunha, M.P., Gabilan, N.H., Rodrigues, A.L., 2012. The role of the NMDA receptors and l-arginine-nitric oxide-cyclic guanosine monophosphate pathway in the antidepressant-like effect of duloxetine in the forced swimming test. *Pharmacol. Biochem. Behav.* 103 (2), 408–417.